A Cheatsheet for World Models

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Chapter 1

Energy-Based Model

This chapter follows [1]. Consider a model with two sets of variables $\begin{cases} X & \text{the observed variables} \\ Y & \text{variables to be predicted} \end{cases}$. Variable X could be a vector containing the pixels from an image of an object. Variable Y could be a discrete variable that represents the possible category of the object. The model is viewed as an energy function which measures the goodness of each possible configuration of X and Y, denoted $E(Y,X)$. The output number can be interpreted as the degree of compatibility between the values of X and Y. Then the problem becomes

$$\min_{Y \in \mathcal{Y}} E(Y,X)$$

where the set $\mathcal{Y}$ needs to be defined. If the size of $\mathcal{Y}$ is too big, it can be impractical to do brute-force search, while if the size is too small, it may not contain the true Y.

The inference procedure means finding the Y that minimises $E(Y,X)$. The best inference procedure to use depends on the internal structure of the model. For example, if $\mathcal{Y}$ is continuous and $E(Y,X)$ is smooth and well-behaved w.r.t. Y, one may use a gradient-based optimisation algorithm. If Y is a collection of discrete variables and the energy function can be expressed as a factor graph, efficient inference procedures for factor graphs can be used.

For decision-making tasks, the system must give the lowest energy to the correct answer. The energies of other answers are irrelevant as long as they are larger. But since energies are uncalibrated, combining two separately trained energy-based models is not straightforward. Calibrating energies in order to permit such combinations can be done in multiple ways, but the only consistent way involves turning the collection of energies for all possible outputs into a normalised probability distribution. For example, through Gibbs distribution

$$P(Y|X) = \frac{e^{-\beta E(Y,X)}}{\int_{y \in \mathcal{Y}} e^{-\beta E(y,X)} dy}$$

where β is an arbitrary positive constant akin to an inverse temperature and the denominator is called the partition function.

1.1 Energy-Based Training

1.1.1 Loss Functions

Training an EBM consists in finding an energy function that produces the best Y for any X. The search for the best energy function is performed within a family of energy functions $\mathscr{E}$ indexed by a parameter W, $\mathscr{E} = \{E(W,Y,X) : W \in \mathscr{W}\}$.

Definition 1.1.1. *The architecture of the EBM is the internal structure of the parametrised energy function $E(W,Y,X)$.*

To train the model, we are given a set of training samples $\mathscr{S} = \{(X^i, Y^i)\}$ where X^i is the input for the ith training sample and Y^i is the corresponding desired answer. In order to find the best energy function in the family $\mathscr{E}$, we need to assess the quality of any particular energy function, based solely on the training set and our prior knowledge about the task. The quality measure is called the loss function, denoted $\mathscr{L}(E,\mathscr{S})$. For simplicity we often denote it $\mathscr{L}(W,\mathscr{S})$ and simply call it the loss function. Then the problem can be re-written as

$$\min_{W \in \mathscr{W}} \mathscr{L}(W,\mathscr{S})$$

In most cases, the loss functional is defined as

$$\mathscr{L}(E,\mathscr{S}) = \frac{1}{P}\sum_{i=1}^{P} L(Y^i, E(W,\mathscr{Y},X^i)) + R(W)$$

where the first term is the average taken over the training set of a per-sample loss function, which depends on Y^i and on the energies obtained by keeping the input sample fixed and varying the answer Y, and $R(W)$ is the regulariser embodying the prior knowledge about which energy functions are preferable to others.

Intuitively, the per-sample loss functional should be designed in such a way that it assigns a low loss to well-behaved energy functions, i.e. energy functions that give the lowest energy to the correct answer and higher energy to all other answers. Conversely, energy functions that do not assign the lowest energy to the correct answers would have a high loss. Training an EBM consists in shaping the energy function, so that for any given X the inference algorithm will produce the desired value for Y. Since the inference algorithm selects the

Y with the lowest energy, the learning procedure must shape the energy surface so that the desired value of Y has lower energy than all other values. Here denote $\begin{cases} Y^i & \text{the correct answer} \\ Y^{*i} & \text{the answer produced by the model} \\ \bar{Y}^i & \text{the most offending incorrect answer} \end{cases}$. With a properly designed loss function, the learning process should have the effect of pushing down on $E(W,Y^i,X^i)$ and pulling up on the incorrect energies, particularly on $E(W,\bar{Y}^i,X^i)$.

e.g.

- energy loss

$$L_{\text{energy}}(W,Y^i,X^i) = E(W,Y^i,X^i)$$

  With this loss function, the wrong answers are not pulled up on and therefore with some architectures, this can lead to a collapsed solution in which the energy is constant and equal to zero. Thus the energy loss will only work with architectures that are designed in such a way that pushing down on $E(W,Y^i,X^i)$ will automatically make the energies of the other answers higher, e.g. $E(W,Y^i,X^i) = ||Y^i - G(W,X^i)||^2$, where G is the regression function.

- generalised perceptron loss

$$L_{\text{perc}}(W,Y^i,X^i) = E(W,Y^i,X^i) - \min_{y \in \mathcal{Y}} E(W,y,X^i)$$

  which is always non-negative since the second term is a lower bound on the first term. Minimising this loss has the effect of pushing down on $E(W,Y^i,X^i)$ while pulling up on the energy of the answer produced by the model, Y^{*i}. Since it has no mechanism for creating an energy gap between the correct answer and the incorrect ones, it may produce flat energy surfaces if the architecture allows it.

- generalised margin loss

  Definition 1.1.2. *Let Y be a $\begin{cases} \text{discrete} \\ \text{continuous} \end{cases}$ variable. Then for a training sample (X^i,Y^i), the most offending incorrect answer $\bar{Y}^i$ is the answer that has the lowest energy among $\begin{cases} \text{all answers that are incorrect} \\ \text{all answers that are at least } \epsilon \text{ away from the correct answer} \end{cases}$;*

$$\bar{Y}^i = \arg\min_{Y \in \mathcal{Y}} E(W,Y,X^i), \quad \begin{cases} Y \neq Y^i \\ ||Y - Y^i|| > \epsilon \end{cases}$$

The generalised margin loss uses the energy of the most offending incorrect answer in the contrastive term

$$L_{\text{margin}}(W,Y^i,X^i) = Q_m(E(W,Y^i,X^i),E(W,\bar{Y}^i,X^i))$$

where m is a positive parameter called the margin and $Q_m(e_1,e_2)$ is a convex function whose gradient has a positive dot product with the vector $[1,-1]$ in the region where $E(W,Y^i,X^i)+m > E(W,\bar{Y}^i,X^i)$, i.e. the loss surface is slanted toward low values of $E(W,Y^i,X^i)$ and high values of $E(W,\bar{Y}^i,X^i)$ wherever $E(W,Y^i,X^i)$ is not smaller than $E(W,\bar{Y}^i,X^i)$ by at least m.

- negative log-likelihood loss

$$L_{\text{nll}}(W,Y^i,X^i) = E(W,Y^i,X^i) + \mathscr{F}_\beta(W,\mathscr{Y},X^i)$$

where $\mathscr{F}$ is the free energy of the ensemble $\{E(W,y,X^i),\ y\in\mathscr{Y}\}$ s.t.

$$\mathscr{F}_\beta(W,\mathscr{Y},X^i) = \frac{1}{\beta}\log(\int_{y\in\mathscr{Y}} \exp(-\beta E(W,y,X^i))dy)$$

where β is a positive constant akin to an inverse temperature.
The form of the nll loss function stems from a probabilistic formulation of the learning problem in terms of the maximum conditional probability principle where observations are independent.

$$\begin{array}{ll} P(Y^1,\ldots,Y^P|X^1,\ldots,X^P,W) = \prod_{i=1}^P P(Y^i|X^i,W) & \\ \leftrightarrow -\log\prod_{i=1}^P P(Y^i|X^i,W) = \sum_{i=1}^P -\log P(Y^i|X^i,W) & \max f \text{ is equivalent to } \min -f \\ = \sum_{i=1}^P (\beta E(W,Y^i,X^i) + \log\int_{y\in\mathscr{Y}} e^{-\beta E(W,y,X^i)}dy) & \text{by the Gibbs distribution} \end{array}$$

Therefore the loss function becomes

$$\mathscr{L}_{\text{nll}}(W,\mathscr{S}) = \frac{1}{P}\sum_{i=1}^P (E(W,Y^i,X^i) + \frac{1}{\beta}\log\int_{y\in\mathscr{Y}} e^{-\beta E(W,y,X^i)}dy)$$

$\rightarrow \frac{\partial L_{\text{nll}}(W,Y^i,X^i)}{\partial W} = \frac{\partial E(W,Y^i,X^i)}{\partial W} - \int_{y\in\mathscr{Y}} \frac{\partial E(W,y,X^i)}{\partial W} P(y|X^i,W)dy$ where $P(Y|X^i,W) = \frac{e^{-\beta E(W,Y,X^i)}}{\int_y e^{-\beta E(W,y,X^i)}dy}$ i.e. the contrastive term causes the energies of all the answers to be pulled up, even the correct answer, in proportion to the likelihood of that answer under the model.

When $\beta\rightarrow\infty$, the NLL loss reduces to the generalised perceptron loss.

1.1.2 Architecture

1.2 Efficient Inference

Chapter 2

VICReg

Chapter 3

The World Model

.1 Graph Laplacian and Spectral Clustering

This appendix follows [2].

.1.1 Preliminaries

Given a set of data points $x_1,\ldots,x_n$ and some notion of similarity $s_{ij} \geq 0$ between all pairs of data points x_i, x_j, the weighted adjacency matrix is $W = (w_{ij})$ where $w_{ij} = \begin{cases} s_{ij} & s_{ij} > t \\ 0 & \text{otherwise} \end{cases}$. Form the similarity graph $G = (V,E)$, where $\begin{cases} v_i & \text{a data point } x_i \\ e_{ij} & \begin{cases} 1 & s_{ij} > t \\ 0 & \text{otherwise} \end{cases} \end{cases}$ where t is a certain threshold. Then the problem of clustering can be reformulated to the problem of partitioning graph s.t. the edges between different groups have very low weights and the edges within a group have high weights. Assume G is undirected, i.e. $W = W^T$, $\forall i,j,\ w_{ij} = w_{ji}$. The degree of a vertex $v_i \in V$ is defined as $d_i = \sum_{j=1}^{n} w_{ij}$. The degree matrix D is defined as the diagonal matrix with degrees $d_1,\ldots,d_n$ on the diagonal. Given a subset of vertices $A \subset V$, $\bar{A} \equiv V \setminus A$. Define the indicator vector $\mathbb{1}_A = (f_1,\ldots,f_n)^T \in \mathbb{R}^n$ as the vector with $f_i = \begin{cases} 1 & v_i \in A \\ 0 & v_i \in \bar{A} \end{cases}$. Also define the size of a subset $A \subset V$, as the number of vertices in A, denoted $|A|$ and $\text{vol}(A) \equiv \sum_{i \in A} d_i$. A subset A is called connected if any two vertices in A can be joined by a path s.t. all intermediate vertices also lie in A. A subset A is called a connected component if it is connected and if there are no connections between vertices in A and $\bar{A}$. For two not necessarily disjoint sets $A, B \subset V$, $W(A,B) \equiv \sum_{i \in A, j \in B} w_{ij}$. The nonempty sets $A_1,\ldots,A_k$ form a partition of the graph if $\begin{cases} A_i \cap A_j = \emptyset \\ A_1 \cup \cdots \cup A_k = V \end{cases}$.

There are several ways to transform a given set $x_1, \ldots, x_n$ of data points with pairwise similarities s_{ij} into a graph. The most widely used ones include the ϵ-neighbourhood graph, k-nearest neighbour graph, and the fully connected graph.

.1.2 Un-normalised Case

The unnormalised graph Laplacian matrix is defined as $L = D - W$.

Proposition .1.1. *The matrix L satisfies the following properties:*

1. $\forall f \in \mathbb{R}^n,\ f^T L f = \frac{1}{2}\sum_{i,j=1}^n w_{ij}(f_i - f_j)^2$
2. *L is symmetric and positive semi definite (p.s.d.).*
3. *The smallest eigenvalue of L is 0, the corresponding eigenvector is the constant one vector $\mathbb{1}$.*
4. *L has n non-negative real-valued eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$*

pf.

- $$\begin{aligned} f^T L f &= f^T D f - f^T W f && \text{by the definition of } L \\ &= \sum_{i=1}^n d_i f_i^2 - \sum_{i,j=1}^n f_i f_j w_{ij} && \text{by the definition of } d_i \\ &= \frac{1}{2}\left(\sum_{i=1}^n d_i f_i^2 - 2\sum_{i,j} f_i f_j w_{ij} + \sum_{j=1}^n d_j f_j^2\right) \\ &= \frac{1}{2}\sum_{i,j} w_{ij}(f_i - f_j)^2 \end{aligned}$$.

- The symmetry of L follows directly from the symmetry of W, D. The p.s.d. is direct consequence of the first stmt, which shows $f^T L f \geq 0,\ \forall f \in \mathbb{R}^n$, given $w_{ij} \geq 0$.

- $L\mathbb{1} = D\mathbb{1} - W\mathbb{1} = (d_i - \sum_{j=1}^n w_{ij})_{i=1,\ldots,n} = 0$ by the definition of d_i, i.e. $\mathbb{1}$ is an eigenvector of L with eigenvalue 0. Since L is p.s.d., all its eigenvalues are non-negative. Thus the smallest eigenvalue of L is 0, with eigenvector $\mathbb{1}$.

- direct consequences of the previous stmts.

Proposition .1.2. *Let G be an undirected graph with non-negative weights. Then the multiplicity k of the eigenvalue 0 of L equals the number of connected components $A_1, \ldots, A_k$ in the graph. The eigenspace of eigenvalue 0 is spanned by the indicator vectors $\mathbb{1}_{A_1}, \ldots, \mathbb{1}_{A_k}$ of those components.*

pf.

- $k = 1$

  the graph is connected. Assume f is an eigenvector with eigenvalue 0. Then $0 = f^T L f = \sum_{ij} w_{ij}(f_i - f_j)^2$ where $w_{ij} \geq 0$, $\forall i, j$. Then it is only possible when $w_{ij}(f_i - f_j)^2$ vanish. Thus if two vertices v_i, v_j are adjacent, $f_i = f_j$. With this argument, f needs to be constant for all vertices which can be connected by a path in the graph. Additionally, as all vertices of a connected component in an undirected graph can be connected by a path, f needs to be constant on the whole connected component. Since we have only 1 component, we only have the constant one vector $\mathbb{1}$ as eigenvector with eigenvalue 0.

- $k \geq 2$

  Wlog, assume that the vertices are ordered according to the connected components they belong to. Then the adjacency matrix W has a block diagonal form and the same is true for the matrix L. Then each of the blocks L_i is a proper graph Laplacian on its own, namely the Laplacian corresponding to the subgraph of the ith connected component. As it is the case for all block diagonal matrices, the spectrum of L is given by the union of the spectra of L_i and the corresponding eigenvectors of L are the eigenvectors of L_i, filled with 0 at the positions of the other blocks. As each L_i is a graph Laplacian of a connected graph, we know each L_i has eigenvalue 0 with multiplicity 1, and the corresponding eigenvector is the constant one vector on the ith connected component. Thus the matrix L has as many eigenvalues 0 as there are connected components and the corresponding eigenvectors are the indicator vectors of the connected components.

We measure the pairwise similarities $s_{ij} = s(x_i, x_j)$ by some similarity function which is symmetric and non-negative and denote the corresponding similarity matrix by $S = (s_{ij})$. → The key trick used here is changing the representation of the abstract data points x_i to points $y_i \in \mathbb{R}^k$. This is possible due to the properties of the graph Laplacian.

.1.3 Spectral Clustering as Graph Cut

The intuition of clustering is to separate points in different groups according to their similarities. For data given in the form of a similarity graph, it can be restated as to find a partition of the graph s.t. the edges between different groups have a very low weight and the edges within a group have high weight.

Algorithm 1 Unnormalised Spectral Clustering

Require: Similarity matrix $S \in \mathbb{R}^{n \times n}$, the number k of clusters to construct.
Ensure: Clusters $A_1, \ldots, A_k$ with $A_i = \{j \mid y_j \in C_i\}$

1: Construct a similarity graph by one of the ways described in Preliminaries. Let W be its weighted adjacency matrix.
2: Compute the unnormalised Laplacian L.
3: Compute the first k eigenvectors $u_1, \ldots, u_k$ of L.
4: Let $U \in \mathbb{R}^{n \times k}$ be the matrix containing the vectors $u_1, \ldots, u_k$ as columns.
5: **for** $i = 1, \ldots, n$ **do**
6: Let $y_i \in \mathbb{R}^k$ be the vector corresponding to the ith row of U.
7: **end for**
8: Cluster the points $(y_i)_{i=1,\ldots,n}$ in $\mathbb{R}^k$ with the k-means algorithm into clusters $C_1, \ldots, C_k$.

e.g. Mincut Problem
For a given number k of subsets, the mincut approach consists in choosing a partition $A_1, \ldots, A_k$ which minimises $\text{cut}(A_1, \ldots, A_k) \equiv \frac{1}{2} \sum_{i=1}^{k} W(A_i, \bar{A}_i)$.
→ when $k = 2$, it can be solved efficiently, but for $k > 2$, in many cases, the solution of mincut simply separates one individual vertex from the rest of the graph.

One solution for the above problem is forcing the sets $A_1, \ldots, A_k$ reasonably large.

- RatioCut

  The size of a subset A of a graph is measured by $|A|$ and the objective function becomes $\text{RatioCut}(A_1, \ldots, A_k) \equiv \frac{1}{2} \sum_{i=1}^{k} \frac{W(A_i, \bar{A}_i)}{|A_i|} = \sum \frac{\text{cut}(A_i, \bar{A}_i)}{|A_i|}$ where $|A_1| = \cdots = |A_k|$ the quantity's minimum is attained.

  - $k = 2$

    $$\min_{A \subset V} \text{RatioCut}(A, \bar{A}), \text{where } A \text{ is a proper subset of } V$$

    Given a nonempty subset $A \subset V$, define the vector $f = (f_1, \ldots, f_n)^T \in \mathbb{R}^n$ where $f_i = \begin{cases} \sqrt{|\bar{A}|/|A|} & v_i \in A \\ -\sqrt{|A|/|\bar{A}|} & v_i \in \bar{A} \end{cases}$. Then the objective function can be rewritten using the unnormalised graph Laplacian

$$f^T L f = \frac{1}{2}\sum_{ij} w_{ij}(f_i - f_j)^2$$

$$= \frac{1}{2}\sum_{i\in A, j\in \bar{A}} w_{ij}\left(\sqrt{\frac{|\bar{A}|}{|A|}} + \sqrt{\frac{|A|}{|\bar{A}|}}\right)^2 + \frac{1}{2}\sum_{i\in \bar{A}, j\in A} w_{ij}\left(-\sqrt{\frac{|\bar{A}|}{|A|}} - \sqrt{\frac{|A|}{|\bar{A}|}}\right)^2 \quad \text{by the definition of } f$$

$$= \mathrm{cut}(A,\bar{A})\left(\frac{|\bar{A}|}{|A|} + \frac{|A|}{|\bar{A}|} + 2\right) \quad \text{by the definition of cut } W(A,\bar{A}) = W(\bar{A},A)$$

$$= \mathrm{cut}(A,\bar{A})\left(\frac{|A|+|\bar{A}|}{|A|} + \frac{|A|+|\bar{A}|}{|\bar{A}|}\right) = |V|\mathrm{RatioCut}(A,\bar{A})$$

Additionally $\sum_i f_i = \sum_{i\in A}\sqrt{\frac{|\bar{A}|}{|A|}} - \sum_{i\in\bar{A}}\sqrt{\frac{|A|}{|\bar{A}|}} = |A|\sqrt{\frac{|\bar{A}|}{|A|}} - |\bar{A}|\sqrt{\frac{|A|}{|\bar{A}|}} = 0$, i.e. the vector f is orthogonal to the constant one vector. Also $||f||^2 = \sum_i f_i^2 = |A|\frac{|\bar{A}|}{|A|} + |\bar{A}|\frac{|A|}{|\bar{A}|} = |\bar{A}| + |A| = n$. Thus the problem can be rewritten as

$$\min_{A\subset V} f^T L f \text{ s.t. } \begin{cases} f \perp \mathbb{1} \\ ||f|| = \sqrt{n} \end{cases}, \text{ where } f_i = \begin{cases} \sqrt{|\bar{A}|/|A|} & v_i \in A \\ -\sqrt{|A|/|\bar{A}|} & v_i \in \bar{A} \end{cases}$$

This is a discrete optimisation problem as the entries of the solution vector f are only allowed to take two particular values and it is NP-hard. So by discarding the discreteness condition, we can get the relaxed optimisation problem

$$\min_{f\in\mathbb{R}^n} f^T L f \text{ s.t. } \begin{cases} f \perp \mathbb{1} \\ ||f|| = \sqrt{n} \end{cases}$$

Then by the Rayleigh–Ritz theorem, the solution of this problem is given by the vector f which is the eigenvector corresponding to the second smallest eigenvalue of L, given that the smallest eigenvalue of L is 0 with eigenvector $\mathbb{1}$. Then we re-transform the real-valued solution vector f of the relaxed problem into a discrete indicator vector. The simplest way is to use the sign of f as indicator function, i.e. $\begin{cases} v_i \in A & f_i \geq 0 \\ v_i \in \bar{A} & f_i < 0 \end{cases}$. But this heuristic is too simple, and most spectral clustering algorithms use $\begin{cases} v_i \in A & f_i \in C \\ v_i \in \bar{A} & f_i \in \bar{C} \end{cases}$ where $C, \bar{C}$ are groups clustering f_i in $\mathbb{R}$.

– arbitrary k

Given a partition of V into k sets $A_1, \ldots, A_k$, we define k indicator vectors $h_j = (h_{1j}, \ldots, h_{nj})^T$ where $h_{ij} = \begin{cases} \frac{1}{\sqrt{|A_j|}} & v_i \in A_j \\ 0 & \text{otherwise} \end{cases}$. Then define the matrix $H \in \mathbb{R}^{n\times k}$ as the matrix containing those k indicator vectors as columns. Then the columns of H are orthonormal, i.e. $H^T H = I$.

Also $h_i^T L h_i = \frac{\text{cut}(A_i, \bar{A}_i)}{|A_i|} = (H^T L H)_{ii}$. This implies $\text{RatioCut}(A_1, \ldots, A_k) = \sum_{i=1}^k h_i^T L h_i = \sum (H^T L H)_{ii} = \text{Tr}(H^T L H)$. So the problem becomes

$$\min_{A_1,\ldots,A_k} \text{Tr}(H^T L H) \text{ s.t. } H^T H = I, \text{ where } h_{ij} = \begin{cases} \frac{1}{\sqrt{|A_j|}} & v_i \in A_j \\ 0 & \text{otherwise} \end{cases}$$

Then we can relax this problem by allowing the entries of the matrix H to take arbitrary real values, and get

$$\min_{H \in \mathbb{R}^{n \times k}} \text{Tr}(H^T L H) \text{ s.t. } H^T H = I$$

By the Rayleigh–Ritz theorem, the solution is given by choosing H as the matrix which contains the first k eigenvectors of L as columns. Here the matrix H is in fact the matrix U used in the unnormalised spectral clustering algorithm. Then we can re-convert the real-valued solution matrix to a discrete partition as previously.

.1.4 Normalised Case

There are two matrices called normalised graph Laplacians:

$$\begin{cases} L_{sym} \equiv D^{-1/2} L D^{-1/2} = I - D^{-1/2} W D^{-1/2} \\ L_{rw} \equiv D^{-1} L = I - D^{-1} W \end{cases}.$$

Proposition .1.3. *Given there is no isolated vertex, the normalised Laplacians satisfy the following properties:*

1. $\forall f \in \mathbb{R}^n,\ f^T L_{sym} f = \frac{1}{2} \sum_{i,j=1}^n w_{ij} (\frac{f_i}{\sqrt{d_i}} - \frac{f_j}{\sqrt{d_j}})^2$

2. *λ is an eigenvalue of L_{rw} with eigenvector u iff λ is an eigenvalue of L_{sym} with eigenvector $w = D^{1/2} u$.*

3. *λ is an eigenvalue of L_{rw} with eigenvector u iff λ and u solve the generalised eigenproblem $Lu = \lambda D u$.*

4. *0 is an eigenvalue of L_{rw} with the constant one vector $\mathbb{1}$ as eigenvector. 0 is an eigenvalue of L_{sym} with eigenvector $D^{1/2}\mathbb{1}$.*

5. *L_{sym} is p.s.d. and L_{sym},L_{rw} have n non-negative real-valued eigenvalues $0 = \lambda_1 \leq \cdots \leq \lambda_n$.*

pf.

- $$\begin{aligned} f^T L_{sym} f &= f^T f - f^T D^{-1/2} W D^{-1/2} f \quad \text{by the definition of } L_{sym} \\ &= \sum_{i=1}^n f_i^2 - \sum_{ij} \frac{f_i}{\sqrt{d_i}} \frac{f_j}{\sqrt{d_j}} w_{ij} \\ &= \frac{1}{2}(\sum f_i^2 - 2\sum_{ij} \frac{f_i f_j}{\sqrt{d_i d_j}} w_{ij} + \sum_j f_j^2) \\ &= \frac{1}{2} \sum_{ij} (\frac{f_j}{\sqrt{d_j}} - \frac{f_i}{\sqrt{d_i}})^2 w_{ij} \quad \text{by the definition of } d_j \end{aligned}$$

- Multiplying the eigenvalue equation $L_{sym} w = \lambda w$ by $D^{-1/2}$ gives $D^{-1} L D^{-1/2} w = \lambda D^{-1/2} w$ which is equivalent to $D^{-1} L u = \lambda u$ where $u = D^{-1/2} w$.
- Multiplying the eigenvalue equation $L_{rw} u = \lambda u$ by D gives $Lu = \lambda D u$.
- $L_{rw} \mathbb{1} = 0$ gives the first stmt. The second stmt follows from stmt 2.
- Since L_{sym} is p.s.d. by stmt 1, its eigenvalues are all non-negative and by an argument similar to the unnormalised case, the stmt about L_{sym} holds. Then by the second stmt, the stmt holds for L_{rw}.
  $\rightarrow L_{rw}$ is not symmetric!

Proposition .1.4. *Let G be an undirected graph with non-negative weights and with no isolated vertex. Then the multiplicity k of the eigenvalue 0 of both L_{rw} and L_{sym} equals the number of connected components $A_1, \ldots, A_k$ in the graph. For L_{rw}, the eigenspace of 0 is spanned by the indicator vectors $\mathbb{1}_{A_i}$ of those components. For L_{sym}, the eigenspace of 0 is spanned by the vectors $D^{1/2} \mathbb{1}_{A_i}$.*

pf. Similar to the unnormalised case.

Algorithm 2 Normalised Spectral Clustering

Require: Similarity matrix $S \in \mathbb{R}^{n \times n}$, the number k of clusters to construct.
Ensure: Clusters $A_1, \ldots, A_k$ with $A_i = \{j \mid y_j \in C_i\}$

1: Construct a similarity graph by one of the ways described in Preliminaries. Let W be its weighted adjacency matrix.
2: Compute the unnormalised Laplacian L.
3: Compute the first k generalised eigenvectors $u_1, \ldots, u_k$ of the generalised eigenproblem $Lu = \lambda D u$.
4: Let $U \in \mathbb{R}^{n \times k}$ be the matrix containing the vectors $u_1, \ldots, u_k$ as columns.
5: **for** $i = 1, \ldots, n$ **do**
6: Let $y_i \in \mathbb{R}^k$ be the vector corresponding to the ith row of U.
7: **end for**
8: Cluster the points $(y_i)_{i=1,\ldots,n}$ in $\mathbb{R}^k$ with the k-means algorithm into clusters $C_1, \ldots, C_k$.

$\rightarrow$ we can replace $Lu = \lambda D u$ with $L_{sym} w = \lambda w$ and get another version of normalised spectral clustering algorithm.

- NCUT

  The size of a subset A is measured by the vol(A) and the objective function becomes $\text{NCut}(A_1,\dots,A_k) \equiv \frac{1}{2}\sum_{i=1}^{k} \frac{W(A_i,\bar{A}_i)}{\text{vol}(A_i)} = \sum \frac{\text{cut}(A_i,\bar{A}_i)}{\text{vol}(A_i)}$ where $\text{vol}(A_1) = \cdots = \text{vol}(A_k)$ the quantity's minimum is attained.

  - $k = 2$

    Define the cluster indicator vector f where $f_i = \begin{cases} \sqrt{\frac{\text{vol}(\bar{A})}{\text{vol}(A)}} & v_i \in A \\ -\sqrt{\frac{\text{vol}(A)}{\text{vol}(\bar{A})}} & v_i \in \bar{A} \end{cases}$.

    Then $\begin{cases} (Df)^T\mathbb{1} = 0 \\ f^TDf = \text{vol}(V) \\ f^TLf = \text{vol}(V)\text{NCut}(A,\bar{A}) \end{cases}$ enable us to rewrite the problem as

    $$\min_A f^TLf \text{ s.t. } \begin{cases} Df \perp \mathbb{1} \\ f^TDf = \text{vol}(V) \end{cases} \text{ where } f_i = \begin{cases} \sqrt{\frac{\text{vol}(\bar{A})}{\text{vol}(A)}} & v_i \in A \\ -\sqrt{\frac{\text{vol}(A)}{\text{vol}(\bar{A})}} & v_i \in \bar{A} \end{cases}$$

    Then we relax the problem by allowing f to take arbitrary real values and the problem becomes

    $$\min_{f\in\mathbb{R}^n} f^TLf \text{ s.t. } \begin{cases} Df \perp \mathbb{1} \\ f^TDf = \text{vol}(V) \end{cases}$$

    $\rightarrow$ substitute f with $g \equiv D^{1/2}f$. Then the problem becomes $\min_{g\in\mathbb{R}^n} g^TD^{-1/2}LD^{-1/2}g$ s.t. $\begin{cases} g \perp D^{1/2}\mathbb{1} \\ ||g||^2 = \text{vol}(V) \end{cases}$, where $\begin{cases} D^{-1/2}LD^{-1/2} = L_{sym} \\ D^{1/2}\mathbb{1} \text{ is the first eigenvector of } L_{sym} \\ \text{vol}(V) \text{ is constant} \end{cases}$. Thus by the Rayleigh–Ritz theorem, its solution g is given by the second eigenvector of L_{sym}. Resubstituting $f = D^{-1/2}g$ and using Proposition .1.3, f is the second eigenvector of L_{rw} or equivalently the generalised eigenvector of the generalised eigenproblem $Lu = \lambda Du$.

  - $k > 2$

    Define the indicator vectors $h_j = (h_{1j},\dots,h_{nj})^T$ where $h_{ij} = \begin{cases} \frac{1}{\sqrt{\text{vol}(A_j)}} & v_i \in A_j \\ 0 & \text{otherwise} \end{cases}$.

    Then set the matrix H as the matrix containing those k indicator vec-

tors as columns. Here $\begin{cases} H^T DH = I \\ h_i^T D h_i = 1 \\ h_i^T L h_i = \text{cut}(A_i, \bar{A}_i)/\text{vol}(A_i) \end{cases}$. By those relations, we can rewrite the problem as

$$\min_{A_1,\dots,A_k} \text{Tr}(H^T LH) \text{ s.t. } H^T DH = I \text{ where } h_{ij} = \begin{cases} \frac{1}{\sqrt{\text{vol}(A_j)}} & v_i \in A_j \\ 0 & \text{otherwise} \end{cases}$$

Relaxing the discrete condition and substituting $T = D^{1/2}H$, we get the problem

$$\min_{T \in \mathbb{R}^{n \times k}} \text{Tr}(T^T D^{-1/2} L D^{-1/2} T) \text{ s.t. } T^T T = I$$

This is the standard trace minimisation problem which is solved by the matrix T which contains the first k eigenvectors of L_{sym} as columns. Re-substituting $H = D^{-1/2}T$ and by Proposition .1.3, we see that the solution H consists of the first k eigenvectors of the matrix L_{rw} or the first k generalised eigenvectors of $Lu = \lambda Du$.

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